Grade 8
Unit 4
Lesson 2 Practice

Name____Answer key_____
Date____________________
Period__________________

Rewrite each expression as an equivalent exponential expression with positive exponents.

1. $\left(\frac{3}{4}\right)^{-3} = \left(\frac{4}{3}\right)^{3}$

2. $\left(-\frac{7}{8}\right)^{-2} = \left(-\frac{8}{7}\right)^{2}$

3. $\left(\frac{-4}{5}\right)^{-4} = \left(\frac{5}{-4}\right)^{4}$

4. $\left(\frac{22}{7}\right)^{-7} = \left(\frac{7}{22}\right)^{7}$

5. Match each expression in Column A to an equivalent expression in Column B.

Column A	Column B
$\left(-\frac{11}{10}\right)^{-3}$	$\left(-\frac{11}{10}\right)^{3}$
$\left(\frac{11}{10}\right)^{-3}$	$\left(\frac{10}{11}\right)^{3}$
$\left(-\frac{10}{11}\right)^{-3}$	$\left(\frac{11}{10}\right)^{3}$
$\left(\frac{10}{11}\right)^{-3}$	$\left(-\frac{10}{11}\right)^{3}$

6. Generate an equivalent expression using the laws of exponents. Express your answer using positive exponents.

$$\left(-\frac{14}{17}\right)^{-2} \cdot \left(-\frac{14}{17}\right)^{-4}$$

$$\left(-\frac{17}{14}\right)^{2} \cdot \left(-\frac{17}{14}\right)^{4} \qquad \left(-\frac{17}{14}\right)^{2+4} = \left(-\frac{17}{14}\right)^{6}$$

For each expression, apply one or more exponent laws rewrite as an equivalent expression with a rational base and positive exponent. Check off the exponent laws (s) you applied to generate the expression. Show your work.

Exponential Expression	Apply Negative Exponent Law	Expanded Form	Equivalent Expression with a Rational Base and Positive Exponent	Exponent Laws Applied
7. $\left(\frac{3}{7}\right)^5 \cdot \left(\frac{3}{7}\right)^{-2}$	$\left(\frac{3}{7}\right)^5 \cdot \left(\frac{7}{3}\right)^2$	$\frac{3}{7} \cdot \frac{3}{7} \cdot \frac{3}{7} \cdot \frac{3}{7} \cdot \frac{3}{7} \cdot \frac{7}{3} \cdot \frac{7}{3}$	$\left(\frac{3}{7}\right)^3$	☐ power of a power ☐ power of a product ☒ product of powers ☐ quotient of powers
8. $\left(\frac{8}{9}\right)^{-4} \div \left(\frac{8}{9}\right)^6$	$\left(\frac{9}{8}\right)^4 \cdot \left(\frac{9}{8}\right)^6$	$\frac{9}{8} \cdot \frac{9}{8} \cdot \frac{9}{8} \cdot \frac{9}{8} \cdot \frac{9}{8} \cdot \frac{9}{8} \cdot \frac{9}{8} \cdot \frac{9}{8} \cdot \frac{9}{8} \cdot \frac{9}{8}$	$\left(\frac{9}{8}\right)^{10}$	☐ power of a power ☐ power of a product ☐ product of powers ☒ quotient of powers
9. $\left(\frac{4}{3} \cdot \frac{9}{10}\right)^{-9}$	$\left(\frac{3}{4} \cdot \frac{10}{9}\right)^9$	$\frac{30}{36} \cdot \frac{30}{36} \cdot \frac{30}{36} \cdot \frac{30}{36} \cdot \frac{30}{36} \cdot \frac{30}{36} \cdot \frac{30}{36} \cdot \frac{30}{36} \cdot \frac{30}{36}$	$\left(\frac{30}{36}\right)^9 = \left(\frac{5}{6}\right)^9$	☐ power of a power ☒ power of a product ☐ product of powers ☐ quotient of powers
10. $\left[\left(\frac{3}{5}\right)^2\right]^{-7}$	$\left(\frac{3}{5}\right)^{-14} = \left(\frac{5}{3}\right)^{14}$	$\frac{5}{3} \cdot \frac{5}{3} \cdot \frac{5}{3} \cdot \frac{5}{3} \cdot \frac{5}{3} \cdot \frac{5}{3} \cdot \frac{5}{3} \cdot \frac{5}{3} \cdot \frac{5}{3} \cdot \frac{5}{3} \cdot \frac{5}{3} \cdot \frac{5}{3} \cdot \frac{5}{3} \cdot \frac{5}{3}$	$\left(\frac{5}{3}\right)^{14}$	☒ power of a power ☐ power of a product ☐ product of powers ☐ quotient of powers